

CaneScan-XAI: An Explainable Ensemble Deep Learning Framework for Automated Sugarcane Leaf Disease Classification

Mahfuz Uddin Ahmed
Dept. of CSE
East West University
Dhaka, Bangladesh
mahfuz17ahmed@gmail.com

Shawna Akter
Dept. of CSE
East West University
Dhaka, Bangladesh
ishawna060@gmail.com

Rafid Bin Taher
Dept. of CSE
East West University
Dhaka, Bangladesh
rafidbintaher2004@gmail.com

Moin Uddin Ahmed
Dept. of CSE
East West University
Dhaka, Bangladesh
moinuddinahmed011905@gmail.com

Abstract— Sugarcane is an important agricultural crop that supports both the sugar and bioenergy industries, supporting the livelihoods of millions of farmers. However, early detection remains a major challenge, as manual disease diagnosis methods are time-consuming, costly, and highly dependent on expert knowledge. This study introduces an explainable deep learning framework for effective sugarcane leaf disease classification using 9,269 images representing 14 disease classes collected from two publicly available Mendeley datasets. The dataset was preprocessed and divided into training, validation, and testing sets with a 70:20:10 ratio. Three pre-trained CNN architectures— Xception, InceptionV3, and DenseNet201 were initially evaluated, followed by the enhancement of the best-performing models using SE and CBAM attention mechanisms. A weighted soft-voting ensemble approach was applied to improve classification performance further, achieving an accuracy of 97.20%. To address the need for model transparency, Grad-CAM was integrated to provide visual explanations of model predictions. Furthermore, a lightweight Streamlit-based application was developed to deliver real-time disease detection with explainable outputs, demonstrating the superior performance of explainable AI in supporting smarter and more sustainable sugarcane farming.

Keywords— *Sugarcane Leaf Disease, Deep Learning, Attention Mechanism, CNN, Ensemble Learning, Grad-CAM, Streamlit*

I. INTRODUCTION

Sugarcane is one of the most important commercial crops in the world and plays a key role in the agricultural economy. Automatic diagnosis of sugarcane leaf disease has been gaining more attention as manual field inspection is time-consuming, requires expert knowledge, and is difficult to apply to large cultivation areas. Recent research has shown that deep learning, transfer learning, attention mechanisms, and deployment-oriented systems can improve disease recognition [1]. However, leaf diseases can significantly reduce crop yield and quality if they are not detected early [2]. Expert visual inspection was the primary basis of the conventional disease detection method, which is labor-intensive, time-consuming, and frequently challenging to use consistently in vast agricultural areas [3].

Recent advances in deep learning have improved the automatic classification of plant diseases, with transfer learning, attention mechanisms, and ensemble learning showing promising results [4]. Despite these developments, most of the existing studies mainly focus on classification accuracy, while less attention has been paid to model

interpretability and practical deployment for real-world agricultural applications.

These difficulties motivate this study to propose an explainable deep learning architecture for classifying sugarcane leaf diseases. First, we evaluated three pre-trained CNN models: Xception, InceptionV3, and DenseNet201. Based on their performance, we further enhanced Xception and InceptionV3 using Squeeze-and-Excitation (SE) and the Convolutional Block Attention Module (CBAM). Next, we created weighted soft-voting ensembles based on the baseline and attention-enhanced models to improve classification performance. To increase the transparency of the prediction process, Grad-CAM was used to highlight the regions of the image that affected each prediction [5]. Finally, a lightweight web application based on Streamlit was developed for real-time disease diagnosis with explainable visualizations.

This is how the rest of the paper is structured. Literature review was placed in section II. Section III outlines the dataset and the suggested methodology. The results of the experiment are described in section IV. The explainable AI is presented in section V. The web app was provided in section VI. Section VII addresses the comparative analysis and discussion segment. Section VIII wraps up the paper and suggest further paths of inquiry.

II. RELATED WORKS

Daphal and Koli [3] developed an attention-based CNN with mobile applications for sugarcane leaf diseases. Their work confirmed practical value of combining disease recognition with a mobile platform. However, the model achieved moderate accuracy, and explainable AI was not addressed.

Das et al. [4] proposed a transfer learning ensemble with a binary classification tree and claimed that it achieved high classification performance. Though effective, the multi-stage design increases model complexity, especially when multiple disease classes are involved. To solve the problem of a small number of training data, Li et al. [6] generated the images of sugarcane leaves using SugarcaneGAN. Their approach improves dataset diversity, but the quality of the generated images can still affect model performance.

Srinivasan et al. [7] evaluated several pre-trained deep learning models on an 11-class sugarcane leaf dataset and reported excellent results with EfficientNet-B7 and DenseNet201. Their work primarily focused on classification

accuracy, not model interpretability or deployment. Similarly, a multi-scale attention-based network was developed by Mauya et al. [8], which achieved high accuracy by combining residual learning and attention mechanisms. However, the study also pointed out issues with the size of datasets and class imbalance. While Nomani et al. [9] performed a comparative analysis of different deep learning models, providing useful performance benchmarks, they do not integrate attention mechanisms or explainable AI.

Zaidal et al. [10] reviewed recent deep learning approaches for sugarcane leaf disease detection and proved that pretrained CNNs and hybrid models remain popular choices. However, since it is a review study, there is no experimental validation. Lavhe et al. [11] developed an IoT-enabled architecture for real-time prediction of diseases and improvement of yield. The system is practical for deployment, but it focuses more on IoT integration than on a comparison of multiple attention-enhanced models with explainable visualizations.

Overall, recent studies confirm that deep learning has become an effective solution for the classification [12] of sugarcane leaf diseases. However, many existing approaches mainly focus on improving accuracy with less attention to model interpretability and practical deployment [13]. Motivated by these limitations, the proposed study evaluates three pretrained CNN models, enhances the two best-performing models with attention mechanisms, combines them through weighted soft-voting ensembles, applies Grad-CAM-based explainability, and develops a lightweight web application for real-time disease diagnosis.

III. METHODOLOGY

The methodology of this study is shown in Fig. 1. The picture shows the overall work done in this research. A dataset was first selected for this study, then the dataset was preprocessed for training, followed by three pre-trained models being used. Attention mechanism was set between the best performed models. SE, along with CBAM, was utilized. After that, an ensemble was applied to merge two models and for best models. XAI was used with the best model for better interpretability. And finally, a prototype web application was also demonstrated in this study.

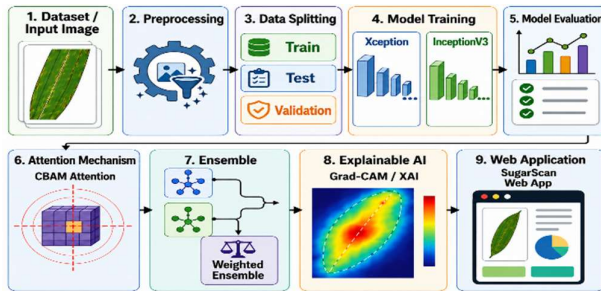


Fig. 1. Workflow of the proposed disease classification framework.

A. Dataset

In this work, the dataset was created by merging two publicly available Mendeley datasets [14][15]. After merging, the final dataset contained 9,269 images belonging to 14 classes: Banded Chlorosis (471), Brown Spot (1,722), Brown Rust (314), Dried Leaves (343), Grassy Shoot (346), Healthy Leaves (952), Mosaic (462), Pokkah Boeng (297), Red Rot (518), Rust (514), Sett Rot (652), Smut (316), Viral

Disease (663), and Yellow Leaf (1,699). The distribution of images in the classes was not consistent, with Brown Spot, Yellow Leaf, and Healthy Leaves having more samples than the remaining classes. This disparity hinders the models from performing effectively. A Class Weight (CW) method was utilized during training to tackle this problem. Fig. 2 displays dataset images for every class.

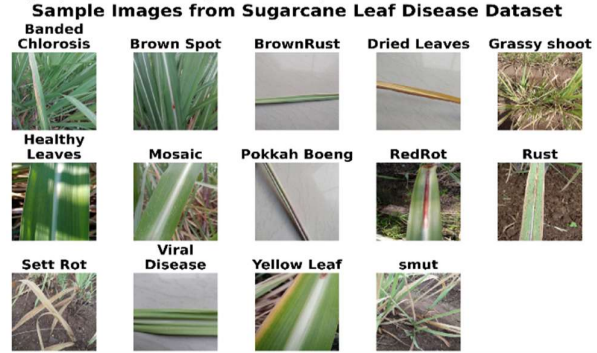


Fig. 2. Dataset images of each class of sugarcane leaf diseases.

B. Preprocessing

Preprocessing involved resizing each image to 224 x 224 pixels. After that, the dataset was split into 70:20:10 ratio for training, validation and testing. This split ensured a fair assessment on unseen data while supporting effective learning and evaluation of the models [16].

C. Pre-Trained Models

Three pre-trained CNN architectures were used as the baseline models for classifying sugarcane leaves.

- Xception: Xception effectively extracts features from images using depth-wise separable convolutions. [17]. It was utilized to identify visual patterns like spots and alterations in leaf color.
- InceptionV3: Using its multi-branch architecture, InceptionV3 captures features at various scales. This makes it effective for recognizing disease symptoms that differ in size and look [18].
- DenseNet201: DenseNet201 employs links between consecutive layers to utilize features learned earlier [19]. It was added as an additional baseline model to assess its feature-learning capability in comparison to Xception and InceptionV3.

D. Attention Mechanism

In order to assess the models' impact on feature learning, two attention modules, Squeeze-and-Excitation (SE) and Convolutional Block Attention Module (CBAM) were applied.

- Squeeze-and-Excitation Block: The SE block emphasizes the significance of separate feature channels. Global average pooling is initially applied to generate a concise representation of the feature maps, which is subsequently processed through fully connected layers to allocate channel-wise weights [20]. These weights improve valuable features and minimize the impact of less significant ones.
- Convolutional Block Attention Module: CBAM enhances the features by employing two forms of attention: channel attention and spatial attention. Channel attention identifies the more valuable features, whereas spatial attention reveals the locations of the relevant information within the image [20].

E. Ensemble Learning

The ensemble method combines several models to enhance performance. Each model produced a probability distribution using SoftMax. The final prediction was calculated using weighted probability averaging.

F. Training Setup

The training configuration played a crucial role in this research, as displayed in Table I. This study utilized the Adam optimizer with a learning rate of 0.001. The size of the batch was 32. For various classes, the suitable loss function, Categorical cross-entropy, was utilized. To minimize overfitting, early stopping was implemented with a patience of 5 [21]. The table provided the number of epochs for each of the models.

TABLE I. TRAINING HYPERPARAMETERS USED FOR ARCHITECTURES

Parameter	Xception	InceptionV3	DenseNet201
Batch Size	32	32	32
Loss function	Categorical cross-entropy	Categorical cross-entropy	Categorical cross-entropy
Learning rate	0.001	0.001	0.001
Optimizer	Adam	Adam	Adam
Epochs	13	27	20
Patience	5	5	5

IV. EXPERIMENTAL RESULTS

This part presents all the experimental results for three pre-trained models used in this work.

A. Xception

The Xception model achieved the best performance among all the models with an accuracy of 94.72%. The training and validation loss curve in Fig. 3 shows a sharp decrease during the early epochs and then gradually stabilizes.

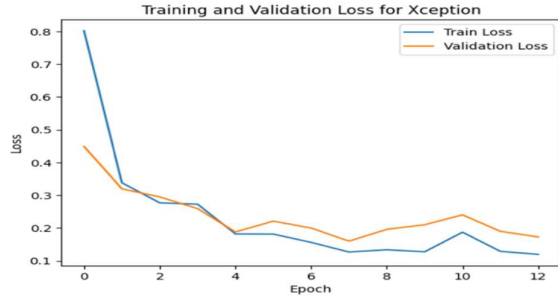


Fig. 3. Training and validation curve of Xception model.

Although a small fluctuation appears near the later epochs, the overall trend indicates stable learning and good generalization.

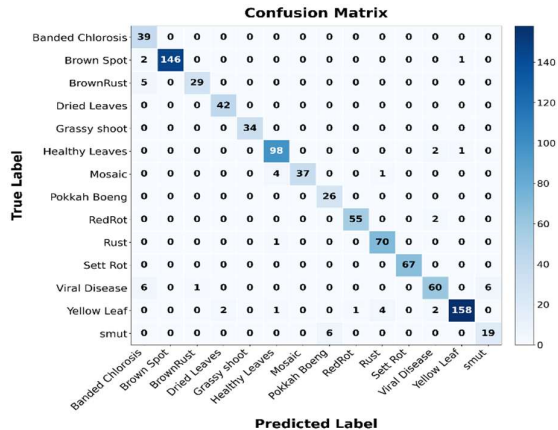


Fig. 4. Confusion matrix of Xception model.

Fig. 4 presents the confusion matrix of the Xception model. Mostly were corrected, as indicated by the high values along the diagonal. Only a few misclassifications were observed between visually similar disease classes, demonstrating that the model learned discriminative features effectively.

B. InceptionV3

The InceptionV3 model obtained an accuracy of 90.62%. From the loss curve in Fig. 5, both training and validation losses decreased rapidly at the beginning, showing that the model learned useful features from the dataset. However, there are several fluctuations in the later epochs, which suggest that the model was less stable compared with the Xception model.

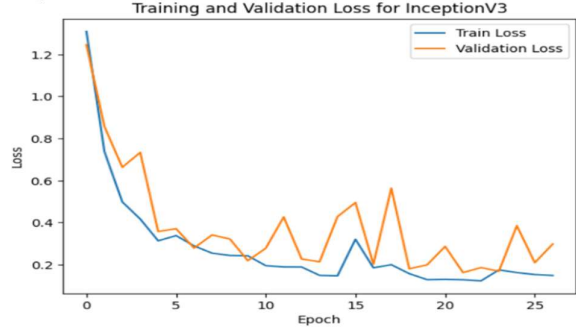


Fig. 5. Training and validation curve for InceptionV3 architectures.

Confusion matrix for the InceptionV3 model was illustrated in Fig. 6. Most disease classes were classified correctly, as reflected by the dominant diagonal values. A few samples were misclassified into visually similar classes, which contributed to the lower overall performance compared with the Xception model.

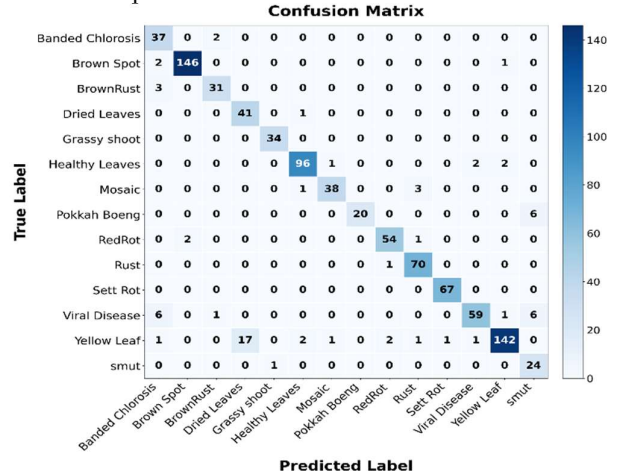


Fig. 6. Confusion matrix of InceptionV3 architectures.

C. DenseNet201

DenseNet201 pre-trained achieved an accuracy of 90.19%. As shown in Fig. 7, the training loss decreased steadily, and the validation loss showed a similar downward trend with small variations. This indicates that the model was able to learn the disease patterns from the leaf images, but the overall result was slightly less than Xception and InceptionV3. The majority of disease types were properly identified, according to the confusion matrix in Fig. 8, even though a limited number of samples were confused with visually similar classes.

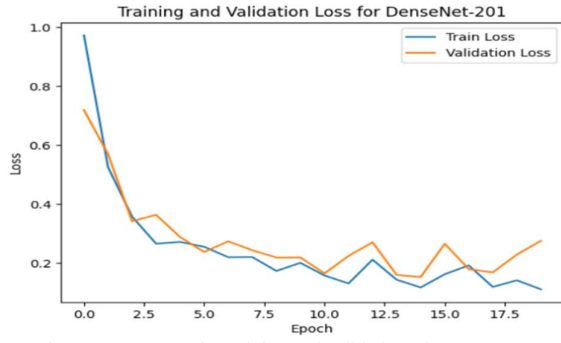


Fig. 7. Loss curve for training and validation of DenseNet201.

Overall, the model produced competitive results but performed slightly below Xception in terms of classification accuracy.

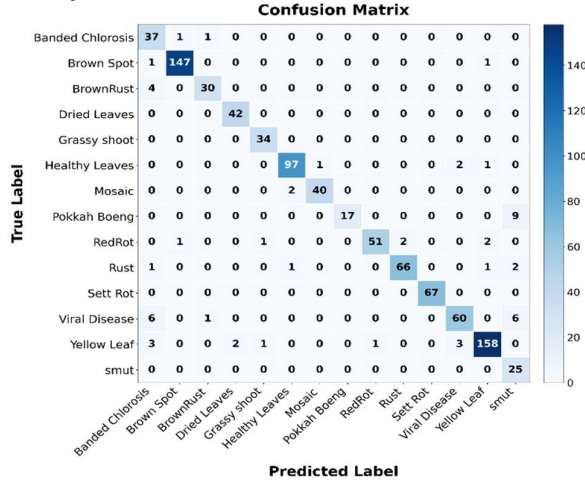


Fig. 8. Confusion matrix of DenseNet201 model.

Table II presents the results of the baseline models used in this work. The models were evaluated using accuracy, precision, recall, and F1-score to identify the best-performing architectures for further experiments.

TABLE II. PERFORMANCE COMPARISON OF THE PRE-TRAINED CNN MODELS

Models	Accuracy (%)	Precision	Recall	F1 Score
Xception	94.72	0.9489	0.9472	0.9464
InceptionV3	90.62	0.9198	0.9062	0.9095
DenseNet201	90.19	0.9201	0.9019	0.9029

Among the three models, Xception achieved the highest accuracy of 94.72%, followed by InceptionV3 (90.62%) and DenseNet201 (90.19%). Since Xception and InceptionV3 showed better overall performance, these two models were selected for further improvement using attention mechanisms.

D. Ensemble Evaluation

After baseline training, Xception and InceptionV3 were selected for attention-based improvement, as they achieved better performance than DenseNet201. Xception with SE block achieved an accuracy of 95.04% while Xception with CBAM block obtained 94.50% accuracy. Applying the SE block with InceptionV3 increased the accuracy to 94.07% and InceptionV3 with the CBAM block gained the best result with an accuracy of 95.15%. After training the attention-based models, ensemble learning was used to improve prediction stability. Three ensemble models were evaluated. Table III summarizes the performance of the three models. The results

show how combining the predictions of Xception and InceptionV3 improved the overall classification performance.

TABLE III. COMPARATIVE PERFORMANCE OF BASELINE, ATTENTION-ENHANCED, AND ENSEMBLE MODELS

Models	SE block	CBAM block	Base model
Xception	95.04%	94.50%	94.72%
InceptionV3	94.07%	95.15%	90.62%
Ensemble model	96.34%	97.20%	97.09%

- **Ensemble 1:** The ensemble of the baseline models achieved 97.09% accuracy, demonstrating that combining the two models improved prediction performance.
- **Ensemble 2:** The ensemble built with the SE-enhanced models achieved 96.34% accuracy.
- **Ensemble 3:** The ensemble of Xception (CBAM) and InceptionV3 (CBAM) achieved the highest accuracy of 97.20% and was selected as the final model of this study.

The contribution is not a new standalone algorithm, but a systematically evaluated explainable framework that combines pretrained CNNs, SE/CBAM attention, weighted ensemble learning, Grad-CAM, and a lightweight web application for 14-class sugarcane disease classification. Fig. 9 shows the confusion matrix of the final CBAM-based ensemble model.

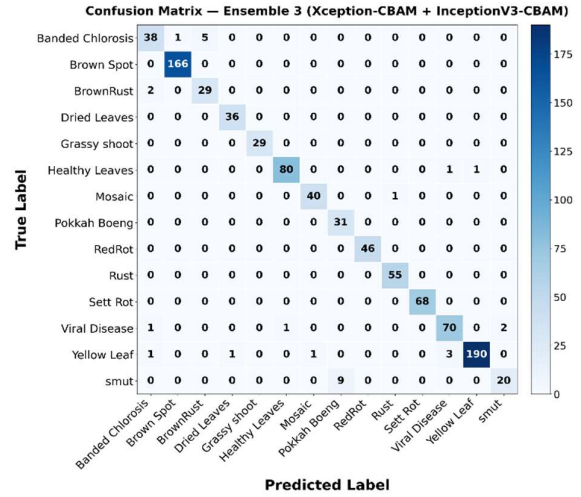


Fig. 9. Confusion matrix of CaneScan-XAI model.

Most of the samples were correctly classified, as indicated by the strong diagonal values. Only a small number of misclassifications were observed between visually similar disease classes, demonstrating the effectiveness of the proposed ensemble model. Fig. 10 shows the training and validation loss curves of the CaneScan-XAI architecture. The curves illustrate the learning behavior of the model throughout the training process.

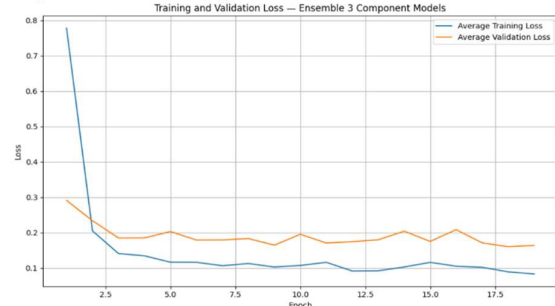


Fig. 10. Loss curve for training and validation of CaneScan-XAI.

Both training and validation losses decreased rapidly during the initial epochs and gradually stabilized, indicating good convergence without significant overfitting.

V. EXPLAINABLE AI

This part shows visual interpretations produced with Grad-CAM. As illustrated in Fig. 11, highlighted regions indicate the image areas that contributed most to the model's prediction, providing a clearer understanding of the decision-making process and improving model interpretability.

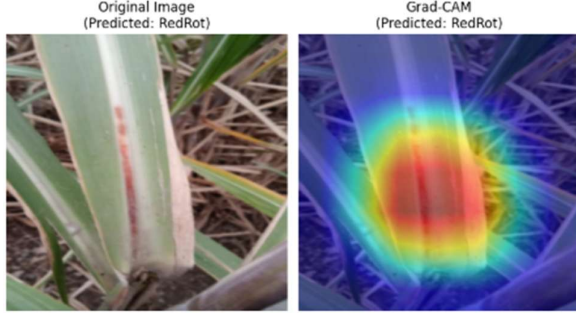


Fig. 11. Grad-CAM visualization highlighting the predicted disease region.

VI. CANESCAN-XAI WEB APP

A lightweight streamlit based web app called CaneScan-XAI was developed to show the usefulness of the suggested architecture, as seen in fig. 12.

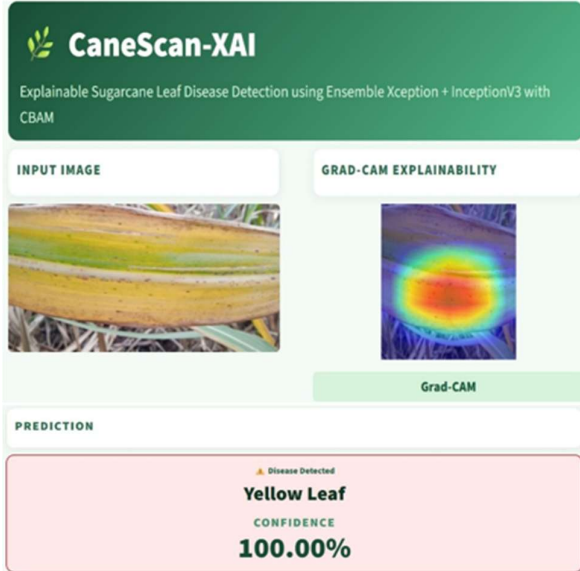


Fig. 12. CaneScan-XAI web app for sugarcane leaf disease.

The app allows users to upload a sugarcane leaf image and predict the disease class using the proposed CaneScan-XAI model. For interpretability, the system provides Grad-CAM visualizations to highlight the important leaf regions [22][23]. Although this application has not been tested in real field conditions.

VII. PERFORMANCE ANALYSIS AND DISCUSSION

A. CNN Architectures Results

Fig. 13 compares the result of the baseline models and their attention-enhanced variants. Among the baseline models, Xception achieved higher accuracy (94.72%) than InceptionV3 (90.62%). After incorporating attention

mechanisms, both models showed improved performance. Xception with the SE block achieved 95.04% accuracy, while InceptionV3 with CBAM reached 95.15%, indicating that attention modules helped the models learn more discriminative features. Among all the evaluated models, the CaneScan-XAI produced the best performance, achieving 97.20% accuracy, along with a precision of 97.37%, a recall of 97.20%, and an F1-score of 97.22%. The best model outperform the others model obtained by the combination of the complementary features of Xception and InceptionV3 with CBAM, which emphasizes the important channels and the disease-related areas. Weighted soft-voting further lowers the model specific errors, resulting the best model.

Model Performance Comparison

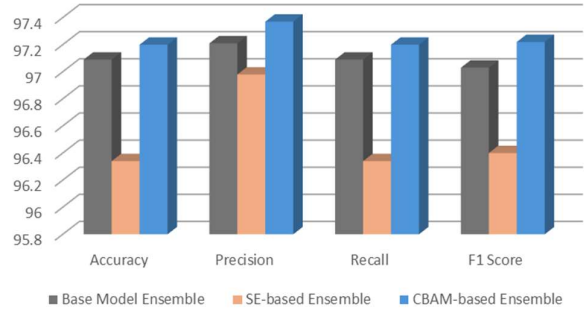


Fig. 13. Performance comparison of baseline, SE, and CBAM-based ensemble models.

B. Benchmarking Against Prior Work

Table IV compares the proposed framework with several recent studies on sugarcane leaf disease classification. Kunduracioglu et al. [2] reported accuracies of 93.39% using EfficientNet-B6. Daphal et al.[3] achieved 86.53% with their attention-based AMRCNN model, while Mauya et al. [8] reported 94.78% using the MADRN architecture. In comparison, the proposed CaneScan-XAI achieved the highest accuracy of 97.20%. Besides improving classification performance, the proposed framework also integrates attention mechanisms, explainable AI, and a lightweight web application, making it more suitable for practical disease diagnosis.

TABLE IV. COMPARISON OF CANESCAN-XAI WITH PREVIOUS WORKS

Author	Method	Accuracy (%)	Dataset	Web Application
[2]	EfficientNet-b6	93.39	1 dataset, 11 classes	No
[3]	AMRCNN	86.53	1 dataset, 5 classes	Yes
[8]	MADRN	94.78	2 datasets, 5 classes	Yes
Proposed Model	Ensemble 1	97.09	2 datasets, 14 classes	Yes
	CaneScan-XAI	97.20		

C. Discussion of Limitations and Future Work

The findings of this research indicate that integrating attention and ensemble learning enhances the classification of sugarcane leaf diseases. The ultimate CaneScan-XAI ensemble outperformed the single models, and Grad-CAM improved the interpretability of predictions by demonstrating the regions of the leaf utilized by the model. The impact of the attention modules varied between the two models. SE provided a slight enhancement with Xception, whereas CBAM showed a more noticeable advancement with

InceptionV3. The majority of the inaccurate predictions happened among disease classes that exhibited similar visual symptoms.

A constraint of this work is that the experiments utilized publicly accessible datasets. These images are unable to include all the conditions present in actual sugarcane fields. Variations in lighting, backdrop, image resolutions, and camera model may consequently influence the predictions. In upcoming research, additional field images may be gathered, and the models can be evaluated under various field conditions. The web application can additionally be enhanced for improved usability by farmers and agriculture experts.

VIII. CONCLUSION

An explainable deep learning method for classifying sugarcane leaf diseases was created in this study. After evaluating three pre-trained CNN models, the two better performing models were used with SE and CBAM attention modules. The CBAM based ensemble of Xception and InceptionV3 produced the best result, with an accuracy of 97.20%, after several ensemble configurations were examined. To make the categorization results easier to understand, the regions that affected the model predictions were displayed using Grad-CAM. Using these visual explanations, A lightweight web application built on Streamlit was also created to showcase real-time disease prediction.

The research still faces certain limitations, primarily concerning the variety of images that are used. The datasets might not completely capture the variations seen in actual sugarcane fields, including disparities in lighting, background, and image clarity. Future efforts may concentrate on gathering additional field images and evaluating the framework in a broader variety of real-world scenarios. The web app can additionally be enhanced for practical application. In summary, the results indicate that integrating transfer learning, attention mechanism, ensemble learning, and visual explanations can be beneficial for diagnosis diseases in sugarcane leaves.

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